

FINANCE CREDIT SCORE

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Introduction

- Use of **statistical techniques** to decide on credit application.
- Adoption of analytical methods for **accuracy** and efficient **credit evaluation**.
- Beyond loan approval, influence on pricing, credit limits, regulatory capital determination, customer relationship management, and targeted marketing.
- **Modern credit scoring** methods has led to **ethical issues** of **fairness** and **transparency**.
- **Regulatory framework** addresses these ethical issues to safeguard consumer rights and maintain integrity of financial system:
 - Basel II
 - IFRS 9
 - Fair Credit Reporting Act



CREDIT SCORE RANGE

https://www.ffcommunity.com/bin/uploads/2023/10/Credit_Score_Range_Infographic_Web.png

Benefits

- Use of **digitalization** along with mobile transactional and online activities resulted in **wealth of data** that CSPs can use it for better decision making.
- **Less cost and time** involved in processing the loan application.
- Availability of various data set help CSPs to develop the predictive model with higher rate of **accuracy**, which enhances **credit risk assessment**.
- Digital data from social media and online activities allows CSPs to offer credit to small businesses and individuals even in case of **no previous credit history**. Also provide better understanding of an individual financial behavior.



<https://www.forbes.com/sites/forbescommunicationscouncil/2021/06/07/why-data-privacy-is-a-human-right-and-what-businesses-should-do-about-it/?sh=215ff5ce4ec3>

- The **integration of various data types** allows for more refined predictions of an individual's future financial behavior.
- The lack of financial history makes CSPs difficult to approve the credit or to evaluate the creditworthiness of an individual or a small business, in order to address this issue, lenders now **use alternative sources of data** like utility bill payments, the frequency and volume of transactions through mobile wallet, and online purchase history help to evaluate potential borrower's reliability.
- **Regulatory guidelines** ensure ethical data management practices and protect the consumer rights in the financial sector.



<https://essentialdata.com/regulatory-compliance-documentation/>

Ethical Theories

Deontology Theory

Holds good as it focuses on the **rules and strict data privacy and security laws**, additionally, it argues that when companies use personal data for evaluation of the credit approval, they must **be transparent and fair** about the evaluation process at each stage.

Utilitarianism Theory

This theory mainly focuses on the outcome, in this case outcome of credit scoring practices. The alternative use of data to increase financial inclusion as long as it does more good than bad. When company makes decision on the credit approval, it should be done with **transparency and fairness**, meaning no one should be left out due to mistake in the algorithm used to make decisions.

Social Contract Theory

This descriptive theory involves the **agreement and understanding** within a society on what is considered as an ethical behavior. This involves **community engagement** to see what people think about the decisions taken or may be related to the design of rules that the society accepts.



<https://researchguides.library.wisc.edu/c.php?g=178198&p=1487671>

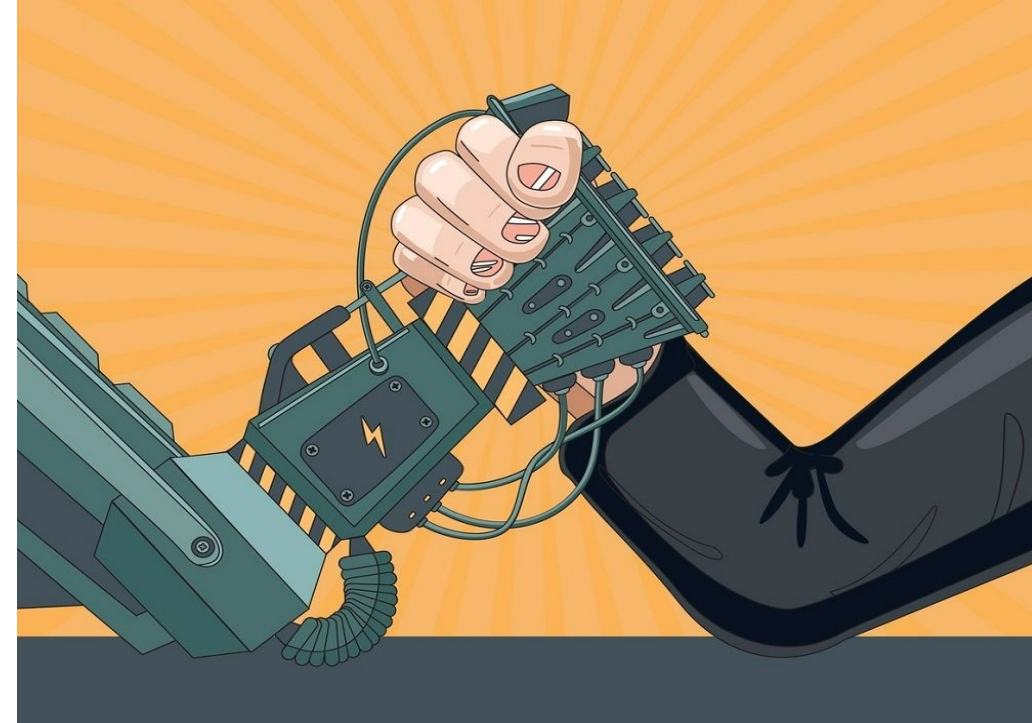
Ethical Issues

- The algorithm could unintentionally **discriminate** against individual based on race, gender etc. as the digital data source includes individual personal information's which are not explicitly included in the data. Th
- Complex algorithms like AI and ML can be **opaque** which makes it difficult to make the consumers and regulatory to understand the process of decision making in the approval or rejection of the credit.
- The challenge is that who should be **accountable** for the data quality, accuracy and fairness of the algorithm as the evaluation of the credit depends upon the third-party data source and the data from algorithm providers.
- There is a high risk of misuse of the digital footprints and personal information as how personal data is collected, used and processed is unknown to the consumers. This is a concern of **data privacy**.
- These ethical issues can be explained well with an example.....



<https://selzy.com/en/blog/ai-ethics/>

- Concern arises on **data security**, as these large volume of data which includes personal details may be at the risk of data breaches, theft and also financial fraud.
- Jurisdiction with less data protection laws, mainly the ethical concern raises on **protecting the consumer rights** and treating the **consumers fairly** is challenging.
- **Uneven application** of credit scoring method due to the technology advancement and legal perspective across countries may cause potential misuse of credit scoring methods.



<https://www.techopedia.com/what-are-some-ethical-issues-regarding-machine-learning/7/33376>

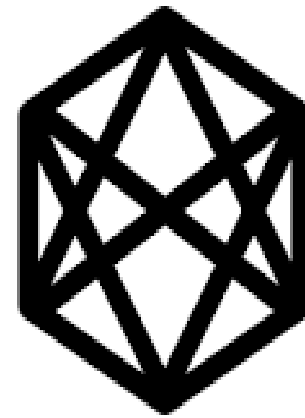
Case Study: Nova Credit

Purpose: Nova Credit help immigrants in USA to use their international credit history to apply for credit domestically.

Stakeholders:

- Immigrants
- U.S Credit Service Providers
- International credit database
- Regulatory bodies that monitors credit reporting

Benefits: Allow immigrants to use their home country credit history to get loans and credit cards in the USA.



**NOVA
CREDIT**

<https://www.novacredit.com/>

Cost:

- Cost of coordinating with the international credit database
- Cost of data security and Privacy compliance
- Cost involved in resolving disputes for incomplete or inaccurate data.

Ethical Concerns:

- Ensuring accuracy and currency of international credit data
- Financial data privacy of individuals across borders
- Concern on discriminatory practices in credit assessment
- Transparency in how the financial private information will be used.



<https://www.russo-consulting.com/informazioni-commerciali/studi/international-credit-report/international-credit-report.htm>

Addressing Ethical Concerns:

- Use of data analytics for regular audit and verifying the accuracy of the credit information
- Adopting data encryption and anonymization to data privacy
- Developing ML algorithm to identify and rectify the biases in credit scoring models
- Being transparent and clearly explaining the criteria of evaluation and what data are collected and how is it been used in US
- Allowing individuals to fill the form of dispute if any of the information is inaccurate & addressing these disputes within 30 days, which ensures fairness and accuracy.

Appropriate Principles, Policies and Practices:

- GDPR (Data minimization; limited purpose; transparency; accountability)
- U.S. Fair Credit Reporting Act (FCRA)
- The Equal Credit Opportunity Act, ECOA (Accuracy; fairness; privacy; no discrimination)
- E2IM (ethical use of AI)



Take Away

- Know about credit systems
- Data Analytics in Finance
- Ethics in data handling
- Challenges in international operations



https://assets4.bigthink.com/system/idea_thumbnails/44964/original/thinking.jpg?1340585574

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<https://link.springer.com/article/10.1007/s00146-023-01676-3>

<https://www.cgap.org/blog/algorithm-bias-in-credit-scoring-whats-inside-black-box>

Thank You